Root-finding decomposition oracles for the point decomposition problem, measured on both tracks

2026-10-12. Study directory `research/pdp_root_finder_20261012/` (README: the full results document; `PROTOCOL.md`: the pre-registration; `reports/run-20261012T2135Z/`: the run report of the timing stage). Requirements and gates: `research/pdp_root_finder_requirements_20261012/` (PR #1734). Driver: `examples/pdp_rootfind/`. **Class: engineering** — every arm's counted work sits on a generic floor (pair table or enumeration); the constants, the exactness instrumentation and the measured distance to F4 are the result; no exponent moved.

What was built and measured

Two mechanisms meet the structural requirements S1–S8 without a Macaulay step: the memoryless **pairs-and-solve** by the group law (binary `rf-pairs`, prime `pairs3`; ≈ 10 multiplications a pair, $|F|^2/2$ pairs a target) and the **2-sum table** (binary `rf-mitm`, prime `mitm3`, `mitm4`; keys $x(P_i \pm P_j)$, $5|F| - 3$ multiplications, one batched inversion and $2|F|$ probes a target, $|F|^2$ keys at 12 bytes). Probes read one 64-byte bucket of sixteen-bit fingerprints (no false negatives, about $16/2^{16}$ false positives confirmed by binary search), all candidates of a target computed first and probed in one prefetched pass. References: PS-0′s `decompose` (`baseline`) and PS-2′s row kernel (`group-scalar`) on the binary rungs, `brute_rational` and the PKM checker's `algebraic_count` on the prime cells, the committed `f4_fp_tower` rows re-read (not re-run), the inherited F4 and the split-dreg F4 on a (23, 8) binary cell.

Cells: PS-0′s frozen corpora $\ell = 8…12$ ($n = 3\ell$, $y^2 + xy = x^3 + x^2 + 1$), fresh corpora at $\ell = 13, 14$ and at $(n, \ell) = (23, 8)$ from the same generator and seed rule, exhaustive digests; the Kummer-tower cells of the PKM rounds at $p_1 = 2013265921$ ($m = 2, 3, 4$), both targets of every committed row, and a fresh ladder $t = 4…13$ ($m = 3$), $4…12$ ($m = 2$), $4…11$ ($m = 4$) with 16 uniform + 8 planted targets a rung.

Exactness (G0)

Binary: verdict and root-set digests equal to the corpus's for both arms on every rung $\ell = 8…14$ (80 targets a rung at $\ell \le 12$, 20 at $\ell = 13, 14$), 0 mismatches; on (23, 8) the verdict digest is equal and 79 of 80 root sets compared equal, the 80th target being degenerate ($x_R \in V$) and routed to the baseline decomposer by construction. Prime: 568 committed rows re-read; 707 / 707 arm root sets equal to the brute-force reference; 479 / 479 finished F4 verdicts equal to the algebraic count (18 unfinished F4 rows excluded); 0 of 486 rows with an algebraic solution and no rational decomposition; every planted target found on every ladder rung; 0 brute-force mismatches at $t \le 7$ ($m = 2, 3$) and $t \le 6$ ($m = 4$); verdicts of all arms equal at every t.

Counted work: the one table

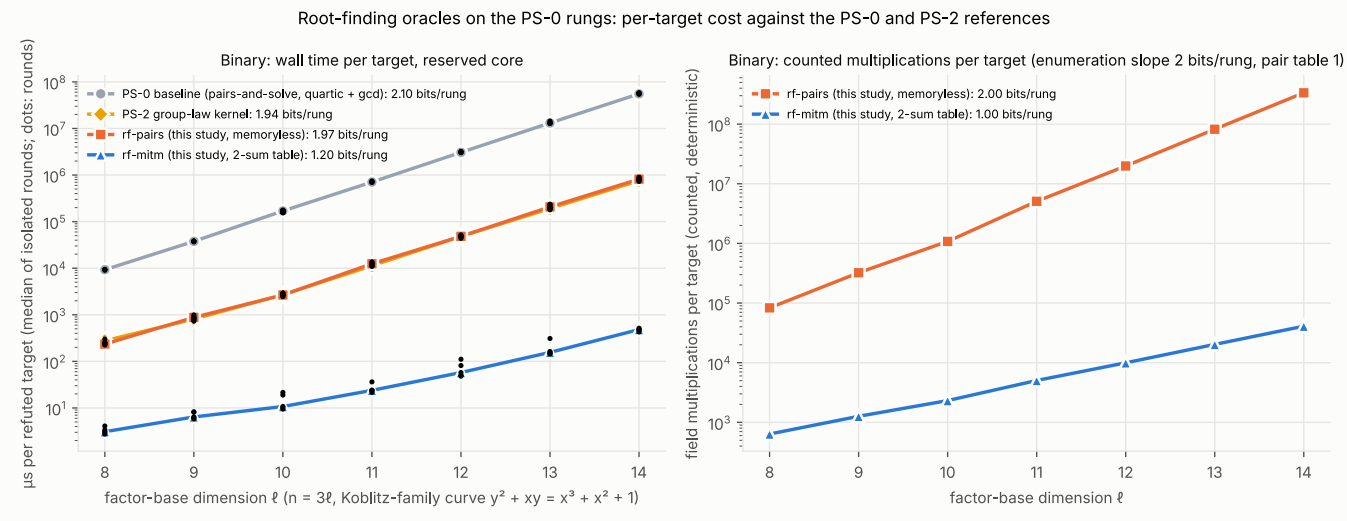


Figure 1. Binary rungs: wall time per refuted target on the reserved core (median of five isolated rounds; dots: rounds) and counted multiplications per target, with the fitted bits per rung. `rf-mitm` is the pair-table floor (1 bit per rung), `rf-pairs` the enumeration floor (2).

Binary, $m = 3$, per refuted uniform target (medians; $T_{\text{enum}} = 2^{2\ell}/2$ candidate pairs, pair floor 2^ℓ probes; wall columns from the five isolated rounds; `baseline` and `group-scalar` carry no counters):

arm	n	ℓ	mul / target	lookups	candidates	÷ T_{enum}	÷ pair floor	µs / target (median)	÷ baseline wall	rounds (contended)	table MB
baseline	24	8	—	—	—	—	—	9,296.9	1.0	4 (0)	—
group-scalar	24	8	—	—	—	—	—	276.5	33.6	2 (0)	—
rf-mitm	24	8	637	256	256	$7.81 \cdot 10^{-3}$	1.00	3.1	3,007.2	5 (0)	0.2
rf-pairs	24	8	83,069	0	33,024	$1.01 \cdot 10^{0}$	—	235.2	39.5	4 (0)	—
baseline	27	9	—	—	—	—	—	37,705.5	1.0	4 (0)	—
group-scalar	27	9	—	—	—	—	—	809.7	46.6	4 (0)	—

rf-mitm	27	9	1,262	506	506	$3.86 \cdot 10^{-3}$	0.99	6.4	5,881.5	5 (0)	0.8
rf-pairs	27	9	322,319	0	128,524	$9.81 \cdot 10^{-1}$	—	879.0	42.9	5 (0)	—
baseline	30	10	—	—	—	—	—	169,656.2	1.0	15 (0)	—
group-scalar	30	10	—	—	—	—	—	2,693.2	63.0	5 (0)	—
rf-mitm	30	10	2,317	928	928	$1.77 \cdot 10^{-3}$	0.91	10.7	15,801.8	5 (0)	2.8
rf-pairs	30	10	1,080,653	0	431,520	$8.23 \cdot 10^{-1}$	—	2,662.9	63.7	5 (0)	—
baseline	33	11	—	—	—	—	—	707,969.2	1.0	5 (0)	—
group-scalar	33	11	—	—	—	—	—	11,290.4	62.7	5 (0)	—
rf-mitm	33	11	5,032	2,014	2,014	$9.60 \cdot 10^{-4}$	0.98	23.9	29,665.5	5 (0)	12.3
rf-pairs	33	11	5,079,305	0	2,030,112	$9.68 \cdot 10^{-1}$	—	12,462.9	56.8	5 (0)	—
baseline	36	12	—	—	—	—	—	3,069,594.3	1.0	5 (0)	—
group-scalar	36	12	—	—	—	—	—	48,094.4	63.8	4 (0)	—
rf-mitm	36	12	9,957	3,984	3,984	$4.75 \cdot 10^{-4}$	0.97	57.6	53,254.7	5 (0)	48.6
rf-pairs	36	12	19,858,245	0	7,940,112	$9.47 \cdot 10^{-1}$	—	48,374.1	63.5	5 (0)	—
baseline	39	13	—	—	—	—	—	13,115,950.2	1.0	5 (0)	—
group-scalar	39	13	—	—	—	—	—	191,078.5	68.6	5 (0)	—
rf-mitm	39	13	20,257	8,104	8,104	$2.42 \cdot 10^{-4}$	0.99	156.0	84,095.5	5 (0)	198.6
rf-pairs	39	13	82,129,985	0	32,845,512	$9.79 \cdot 10^{-1}$	—	206,909.0	63.4	5 (0)	—
baseline	42	14	—	—	—	—	—	55,506,122.0	1.0	5 (0)	—
group-scalar	42	14	—	—	—	—	—	757,478.1	73.3	5 (0)	—
rf-mitm	42	14	40,922	16,370	16,370	$1.22 \cdot 10^{-4}$	1.00	482.0	115,160.1	5 (0)	804.9
rf-pairs	42	14	335,044,787	0	134,004,820	$9.98 \cdot 10^{-1}$	—	818,311.5	67.8	5 (0)	—
baseline	23	8	—	—	—	—	—	8,698.1	1.0	5 (0)	—
group-scalar	23	8	—	—	—	—	—	213.6	40.7	5 (0)	—
rf-mitm	23	8	592	238	238	$7.26 \cdot 10^{-3}$	0.93	3.9	2,230.3	5 (0)	0.2
rf-pairs	23	8	71,873	0	28,560	$8.72 \cdot 10^{-1}$	—	202.2	43.0	5 (0)	—

Prime, Kummer towers at p_1, full scan of 8 uniform targets (the cost of a refutation; $T_{\mathrm{enum}(m,t)} = 2^{(m-1)t}/(m-1)!$, pair floor $2^{\lfloor m/2 \rfloor t}$):

m	arm	t	N	mul / target	candidates	$\div\, T_{\mathrm{enum}}$	÷ pair floor	µs / full scan (median)	rounds (contended)	table MB
2	subtract2	4	8	22	10	$6.25 \cdot 10^{-1}$	0.62	0.2	5 (1)	—
2	subtract2	5	10	67	28	$8.75 \cdot 10^{-1}$	0.88	0.4	5 (1)	—
2	subtract2	6	12	217	88	$1.38 \cdot 10^{0}$	1.38	1.1	5 (1)	—
2	subtract2	7	14	372	150	$1.17 \cdot 10^{0}$	1.17	1.7	5 (1)	—
2	subtract2	8	16	567	228	$8.91 \cdot 10^{-1}$	0.89	2.6	5 (1)	—
2	subtract2	9	18	1,237	496	$9.69 \cdot 10^{-1}$	0.97	5.8	5 (1)	—
2	subtract2	10	20	2,582	1,034	$1.01 \cdot 10^{0}$	1.01	12.6	5 (1)	—
2	subtract2	11	22	5,067	2,028	$9.90 \cdot 10^{-1}$	0.99	34.3	5 (1)	—
2	subtract2	12	24	10,392	4,158	$1.02 \cdot 10^{0}$	1.02	61.7	5 (1)	—
3	mitm3	4	12	32	14	$1.09 \cdot 10^{-1}$	0.88	0.3	5 (1)	0.0
3	mitm3	5	15	77	32	$6.25 \cdot 10^{-2}$	1.00	0.6	5 (1)	0.0
3	mitm3	6	18	167	68	$3.32 \cdot 10^{-2}$	1.06	1.2	5 (1)	0.0
3	mitm3	7	21	302	122	$1.49 \cdot 10^{-2}$	0.95	1.9	5 (1)	0.0
3	mitm3	8	24	632	254	$7.75 \cdot 10^{-3}$	0.99	3.7	5 (1)	0.2
3	mitm3	9	27	1,207	484	$3.69 \cdot 10^{-3}$	0.95	7.0	5 (1)	0.7
3	mitm3	10	30	2,597	1,040	$1.98 \cdot 10^{-3}$	1.02	17.0	5 (1)	4.3
3	mitm3	11	33	5,097	2,040	$9.73 \cdot 10^{-4}$	1.00	34.7	5 (1)	12.5
3	mitm3	12	36	10,147	4,060	$4.84 \cdot 10^{-4}$	0.99	90.4	5 (1)	49.8
3	mitm3	13	39	20,417	8,168	$2.43 \cdot 10^{-4}$	1.00	306.5	5 (1)	200.7
3	pairs3	4	12	305	112	$8.75 \cdot 10^{-1}$	—	2.3	5 (1)	0.0

3	pairs3	5	15	1,421	544	$1.06 \cdot 10^{0}$	—	10.7	5 (1)	0.0
3	pairs3	6	18	6,083	2,380	$1.16 \cdot 10^{0}$	—	46.4	5 (1)	0.0
3	pairs3	7	21	19,151	7,564	$9.23 \cdot 10^{-1}$	—	161.3	5 (1)	0.0
3	pairs3	8	24	81,785	32,512	$9.92 \cdot 10^{-1}$	—	549.5	5 (1)	0.2
3	pairs3	9	27	294,995	117,612	$8.97 \cdot 10^{-1}$	—	2,033.7	5 (1)	0.7
3	pairs3	10	30	1,356,677	541,840	$1.03 \cdot 10^{0}$	—	9,030.2	5 (1)	4.3
3	pairs3	11	33	5,211,177	2,082,840	$9.93 \cdot 10^{-1}$	—	34,147.5	5 (1)	12.5
3	pairs3	12	36	20,622,767	8,245,860	$9.83 \cdot 10^{-1}$	—	135,260.5	5 (1)	49.8
3	pairs3	13	39	83,432,033	33,366,280	$9.94 \cdot 10^{-1}$	—	574,094.3	5 (1)	200.7
4	mitm4	4	16	495	128	$1.88 \cdot 10^{-1}$	0.50	5.5	5 (1)	0.0
4	mitm4	5	20	1,818	450	$8.24 \cdot 10^{-2}$	0.44	17.5	5 (1)	0.0
4	mitm4	6	24	9,608	2,312	$5.29 \cdot 10^{-2}$	0.56	96.6	5 (1)	0.0
4	mitm4	7	28	25,358	6,050	$1.73 \cdot 10^{-2}$	0.37	248.6	5 (1)	0.0
4	mitm4	8	32	127,800	30,258	$1.08 \cdot 10^{-2}$	0.46	1,257.0	5 (1)	0.2
4	mitm4	9	36	595,193	140,450	$6.28 \cdot 10^{-3}$	0.54	5,574.1	5 (1)	1.1
4	mitm4	10	40	2,147,310	506,018	$2.83 \cdot 10^{-3}$	0.48	18,900.8	5 (0)	3.1
4	mitm4	11	44	9,615,903	2,264,192	$1.58 \cdot 10^{-3}$	0.54	94,566.6	5 (0)	17.5

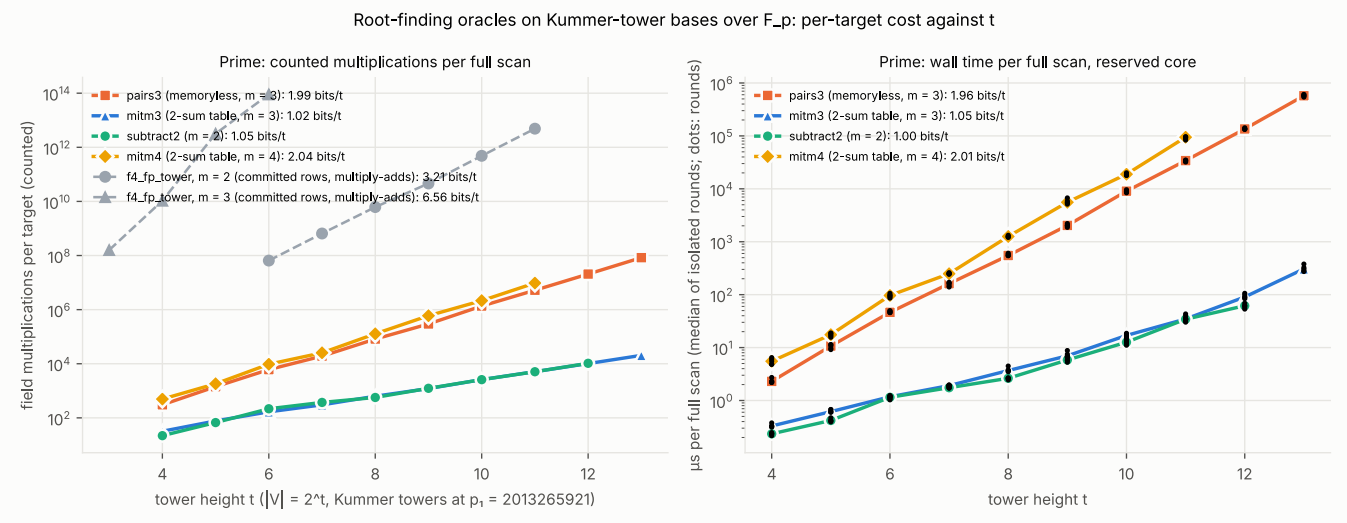


Figure 2. Prime ladder: counted multiplications per full scan against t with the committed F4 rows (multiply-adds, finished runs) as context, and wall time per full scan on the reserved core.

Amortised accounting (G5), from the recorded build counts: the 2-sum table costs about 5 multiplications a key, half of what the memoryless arm spends on one target, so it pays for itself at $K = 1$: binary $\ell = 13$, $T_{\text{prepare}} = 41,032,579$ mul and $T_{\text{target}} = 20,257$, i.e. $T_{\text{prepare}}/K + T_{\text{target}} = 4.1 \cdot 10^{7}, 2.58 \cdot 10^{6}, 1.81 \cdot 10^{5}$ at $K = 1, 16, 256$ against $8.21 \cdot 10^{7}$ for `rf-pairs`; prime $t = 13$: $4.17 \cdot 10^{7}$ to build against $8.34 \cdot 10^{7}$ for `pairs3`.

Against F4 (G2)

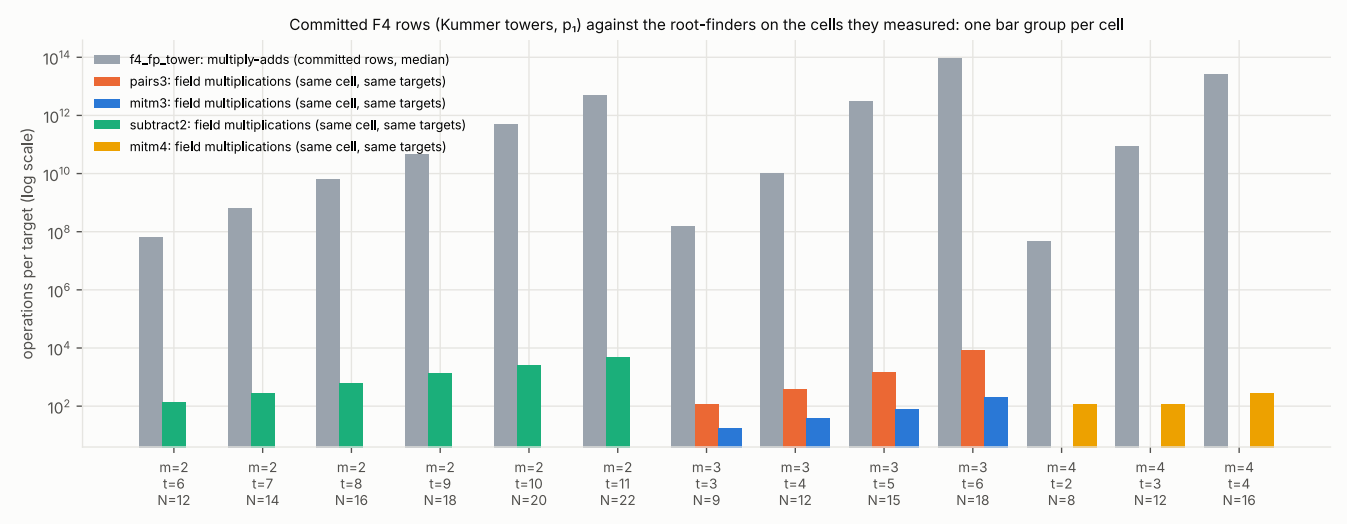


Figure 3. The committed `f4_fp_tower` rows (Kummer, p_1; multiply-adds, finished runs, medians of the rows of a cell) against the root-finders on the same cells and targets (field multiplications).

Prime, same cells and targets as the committed rows (finished F4 runs only; a multiply-add and a multiplication compared one to one, which flatters F4):

m	t	N	finished rows	unfinished	F4 multiply-adds	F4 ms	arm	mul / target	F4 multiply-adds ÷ mul
2	6	12	2	0	$6.50 \cdot 10^{7}$	53.0	subtract2	137	$4.75 \cdot 10^{5}$
2	7	14	2	0	$6.51 \cdot 10^{8}$	225.1	subtract2	277	$2.35 \cdot 10^{6}$
2	8	16	2	0	$6.18 \cdot 10^{9}$	1,110.7	subtract2	632	$9.78 \cdot 10^{6}$
2	9	18	2	0	$4.60 \cdot 10^{10}$	5,961.5	subtract2	1,402	$3.28 \cdot 10^{7}$
2	10	20	2	2	$4.84 \cdot 10^{11}$	56,800.3	subtract2	2,572	$1.88 \cdot 10^{8}$
2	11	22	2	2	$4.88 \cdot 10^{12}$	596,937.0	subtract2	4,862	$1.00 \cdot 10^{9}$
3	3	9	2	0	$1.59 \cdot 10^{8}$	89.8	mitm3	17	$9.36 \cdot 10^{6}$
3	3	9	2	0	$1.59 \cdot 10^{8}$	89.8	pairs3	113	$1.41 \cdot 10^{6}$
3	4	12	2	0	$1.06 \cdot 10^{10}$	1,886.0	mitm3	37	$2.86 \cdot 10^{8}$
3	4	12	2	0	$1.06 \cdot 10^{10}$	1,886.0	pairs3	389	$2.72 \cdot 10^{7}$
3	5	15	2	0	$3.19 \cdot 10^{12}$	370,704.7	mitm3	77	$4.14 \cdot 10^{10}$
3	5	15	2	0	$3.19 \cdot 10^{12}$	370,704.7	pairs3	1,421	$2.24 \cdot 10^{9}$
3	6	18	2	5	$8.99 \cdot 10^{13}$	10,967,967.0	mitm3	197	$4.56 \cdot 10^{11}$
3	6	18	2	5	$8.99 \cdot 10^{13}$	10,967,967.0	pairs3	8,357	$1.08 \cdot 10^{10}$
4	2	8	2	0	$4.84 \cdot 10^{7}$	44.6	mitm4	113	$4.29 \cdot 10^{5}$
4	3	12	3	0	$8.55 \cdot 10^{10}$	9,382.6	mitm4	113	$7.56 \cdot 10^{8}$
4	4	16	2	3	$2.70 \cdot 10^{13}$	3,334,164.6	mitm4	270	$1.00 \cdot 10^{11}$

G2(a) asks for $\leq 0.1\times$; the ratios are 10^{5}–10^{12} and grow with the cell. P4 held on every clause (`mitm3` 197 and `pairs3` 8,357 multiplications at $m = 3, N = 18$ against $8.99 \cdot 10^{13}$; `mitm4` 270 at $m = 4, N = 16$ against $2.71 \cdot 10^{13}$; every $m \geq 3, N \geq 15$ ratio above 10^{9}). G2(b): the finished F4 rows grow by 3.214 ± 0.063 bits per unit of t at $m = 2$ and 6.556 ± 0.864 at $m = 3$ (1.61 and 2.19 bits per tower variable, the figures the requirements document quoted), the root-finders by 1.05 ± 0.09, 1.02 ± 0.02 and 1.99 ± 0.03: disjoint intervals over ≥ 4 sizes on the prime track. Binary (23, 8): the inherited F4 decision oracle costs $4.10 \cdot 10^{7}$ word XORs and 231.5 ms a target (medians of 8, shared host) where `rf-mitm` costs 592 multiplications and 3.9 µs (isolated); the split-dreg F4 did not finish (killed, then an allocation failure under a 12 GB cap: censored). G2(c), the external tools on exported tower systems, was not run.

Growth and the generic line (G3, G5)

track	arm	quantity	bits per rung	± 95 %	r^2	sizes
binary	baseline	ns	2.096	0.011	1.0000	7
binary	group-scalar	ns	1.935	0.077	0.9979	7
binary	rf-mitm	mul	1.005	0.020	0.9995	7
binary	rf-mitm	ns	1.196	0.123	0.9864	7

binary	rf-pairs	mul	2.004	0.040	0.9995	7
binary	rf-pairs	ns	1.973	0.060	0.9988	7
prime m=2	subtract2	mul	1.052	0.089	0.9871	9
prime m=2	subtract2	ns	0.998	0.070	0.9911	9
prime m=3	mitm3	mul	1.019	0.020	0.9992	10
prime m=3	mitm3	ns	1.054	0.089	0.9855	10
prime m=3	pairs3	mul	1.990	0.026	0.9996	10
prime m=3	pairs3	ns	1.962	0.031	0.9995	10
prime m=4	mitm4	mul	2.035	0.061	0.9986	8
prime m=4	mitm4	ns	2.009	0.071	0.9980	8
prime m=2	f4_fp_tower dickson p=786433	muladds	3.792	0.313	0.9878	9
prime m=2	f4_fp_tower dickson p=786433	ms	1.696	0.169	0.9821	9
prime m=2	f4_fp_tower isogeny p=786433	muladds	3.646	0.290	0.9870	10
prime m=2	f4_fp_tower isogeny p=786433	ms	1.819	0.179	0.9803	10
prime m=2	f4_fp_tower kummer p=786433	muladds	3.697	0.273	0.9888	10
prime m=2	f4_fp_tower kummer p=786433	ms	1.849	0.196	0.9772	10
prime m=2	f4_fp_tower kummer p=2013265921	muladds	3.214	0.063	0.9996	6
prime m=2	f4_fp_tower kummer p=2013265921	ms	2.676	0.269	0.9896	6
prime m=3	f4_fp_tower kummer p=2013265921	muladds	6.556	0.864	0.9910	4
prime m=3	f4_fp_tower kummer p=2013265921	ms	5.831	0.840	0.9893	4
prime m=2	f4_fp_tower kummer p=3221225473	muladds	3.927	0.355	0.9874	8
prime m=2	f4_fp_tower kummer p=3221225473	ms	1.780	0.194	0.9818	8

Every root-finder's slope equals its floor within ± 1.96 s.e. (P3, P5). The table arm's ST^2 gap to the generic line ($S = |F|^2$ keys, $T = (\mathrm{mul} + \mathrm{sqr})/6$ group-operation equivalents, $r = 2^{n-1}$, $|\mathrm{Aut}| = 2$) is positive and growing — that line prices a whole DLP, and a decomposition oracle alone cannot pass it; the row is the G5 accounting:

n	ℓ	S (keys)	T (group-op eq.)	$\log_2(ST^2)$	$\log_2(r/\lvert\mathrm{Aut}\rvert)$	gap (bits)
23	8	$1.42 \cdot 10^4$	138.3	28.0	21.0	+7.0
24	8	$1.64 \cdot 10^4$	148.8	28.4	22.0	+6.4
27	9	$6.40 \cdot 10^4$	294.7	32.4	25.0	+7.4
30	10	$2.15 \cdot 10^5$	540.8	35.9	28.0	+7.9
33	11	$1.01 \cdot 10^6$	1,174.3	40.3	31.0	+9.3
36	12	$3.97 \cdot 10^6$	2,323.5	44.3	34.0	+10.3
39	13	$1.64 \cdot 10^7$	4,726.8	48.4	37.0	+11.4
42	14	$6.70 \cdot 10^7$	9,548.7	52.4	40.0	+12.4

G3 fails; nothing here moves an exponent. G6 (end-to-end budgets) follows: with the floors as the measured slopes, the 2-sum oracle at ECC2K-130, $m = 4$ ($|F| \approx 2^{33.9}$) probes $\approx 2^{67.8}$ keys a target against a budget of 2^{35}; at $m = 5$, 2^{81} against 2^{41}; at a 256-bit prime, $m = 4$, $t = 64$: 2^{126} against 2^{59}. Extrapolation, not measurement.

Wall time (reserved core, five rounds)

Medians of the uncontended rounds (`raw/bin_time.jsonl`, `raw/prime_time.jsonl`, `raw/isolated.jsonl`; 177 of 185 commands ran, 8 were refused by the settle check and leave fewer rounds on six $\ell \le 12$ rows, 7 runs flagged contended are excluded). Binary, per refuted target: `rf-mitm` 3.1 µs at $\ell = 8$, 57.6 µs at $\ell = 12$, 156 µs at $\ell = 13$, 482 µs at $\ell = 14$; PS-0′s `baseline` 9.3 ms, 3.07 s, 13.1 s, 55.5 s (3,007×, 53,255×, 84,095×, 115,160×); PS-2′s `group-scalar` 276 µs, 48.1 ms, 191 ms, 757 ms (89×, 834×, 1,225×, 1,572×); `rf-pairs` within 0.91–1.18× of `group-scalar` on every rung (P2). The (23, 8) cell: 8.7 ms, 214 µs, 202 µs, 3.9 µs. `rf-mitm`'s wall grows by 1.196 ± 0.123 bits per rung against 1.000 in counted work: the per-probe cost rises from 12 ns ($\ell = 8$, 0.2 MB table) to 29 ns ($\ell = 14$, 805 MB, past the 260 MiB L3); the table build is recorded apart (8.2 s at $\ell = 14$) and charged in §4. Prime, per full scan: `mitm3` 0.33 µs at $t = 4 \longrightarrow$ 306.5 µs at $t = 13$, `pairs3` 2.3 µs $\longrightarrow$ 574 ms (1,873× apart at $t = 13$), `subtract2` 0.23 $\longrightarrow$ 61.7 µs ($t = 12$), `mitm4` 5.5 µs $\longrightarrow$ 94.6 ms ($t = 11$); wall slopes 1.054 ± 0.089, 1.962 ± 0.031, 0.998 ± 0.070, 2.009 ± 0.071 bits per unit of t. A/A control (the baseline against itself at $\ell = 10$, every round): 155–177 ms a target, at most 7.5 % apart within a pair. P1′s wall clauses ($> 50\times$ at $\ell = 8$, $> 1,000\times$ at $\ell = 12$) held with 3,007× and 53,255×. Against the Gröbner references in wall: inherited F4 at (23, 8) 231.5 ms a target (shared host) against 3.9 µs, about $5.9 \cdot 10^4$ (unlike measurements, an order of magnitude); the committed $m = 3$, $N = 18$ F4 rows 10,968 s on their host against a 1.2 µs full scan here.

Gates read, and status

gate	result	where
G0 exactness	pass on every arm and cell	README §3
G1 vs enumeration	pass for the table arms (by memory, not algebra: ratio $7.8 \cdot 10^{-3} \to 1.2 \cdot 10^{-4}$ binary, $0.11 \to 2.4 \cdot 10^{-4}$ prime $m = 3$, halving a rung); fail (ratio ≈ 1) for the memoryless arms	§4
G2(a) vs F4, counted	pass: prime 10^5–10^{12}; binary: the Gröbner arm that finishes is about $6 \cdot 10^4 \times$ slower in wall, the other does not finish	§5
G2(b) vs F4, growth	pass on the prime track ($m = 2, 3$, disjoint intervals over ≥ 4 sizes); binary: F4′s side is a two-size ratio, not a fit	§5–6
G2(c) vs msolve	not run	—
G3 exponent	fail: every slope is its floor, ST^2 gap positive	§6
G4 vs SAT	not run	—
G5 accounting	reported	§4, §6
G6 end-to-end	fail (extrapolation at the floors)	§8
G7 class	engineering on every row	—

Deviations from the protocol (README §2): the Gröbner reference cell is (23, 8), not (24, 8) (the F4 drivers refuse $n = 24$); the split-dreg reference is censored (memory); prime timing is the full scan of 8 uniform targets at every t (at $t \geq 12$ every uniform target decomposes); binary timing at $\ell \geq 13$ uses 4 refuted targets and one pass; the lookup structure changed before the timing stage with counted units unchanged and exactness repeated; slope intervals are ± 1.96 s.e. over rungs (a bootstrap over targets has zero width for a table arm); the ladders and the committed rows read are supersets of the registered ones; R2, R4, R5, R8 not run.

On the request's wording ("the fastest algorithm you can make"): within the arms measured here the 2-sum table is the fastest on every cell, in counted work and in wall time, by the factors above; its exponent is the known pair-table floor, and nothing measured here bears on the end-to-end sizes. Claims stop there.

Environment: Claude Code cloud container, Intel Xeon @ 2.10 GHz, 4 vCPU, L3 260 MiB, 15 GiB RAM, kernel 6.18.44, rustc 1.97.0, base commit `cd0286878`; timing through `tools/isolated_bench.py run --cpus 3` (CPU 3 reserved, other threads evicted, contended runs listed and not pooled), `RAYON_NUM_THREADS = 1`. Conductor's control plane was unreachable in this container (no Docker daemon or Postgres): no task card attached. Python lays out recorded values only; every table above is generated from `ANALYSIS.json`, written by the Rust `analyze` subcommand from the raw records. Canonical graphs checked and left unchanged: the IC scoreboard, `progress-timeline.json`, the leaderboard and `docs/bounds/FRONTIER.md` (no end-to-end S).